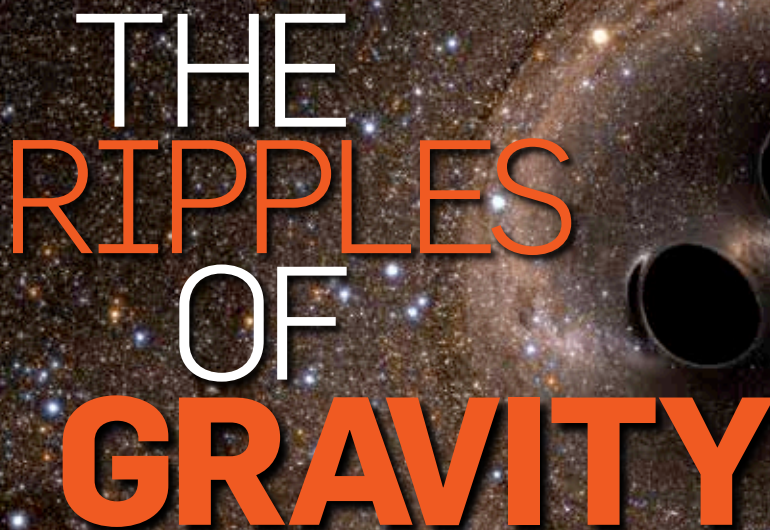




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Space Age



THE RIPPLES OF GRAVITY

The discovery of gravitational waves is of immeasurable significance to our better understanding of the universe and beyond

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[illegible]

it has taken humanity this long to finally construct an experiment intricate enough to capture the data necessary to validate Einstein's predictions. So, one would expect that this would be an extremely complicated experiment that would take a couple of PhD's and a superhuman IQ to even begin to understand. But, at the fundamental level, this is an experiment that can be replicated at home! In this article, we demystify the physics that goes behind detecting gravitational waves and the seminal work that goes behind it.

To understand why gravitational waves are important, one needs to go back in time to when Einstein first proposed his two theories of relativity. The Special Theory of Relativity, first explained in the paper 'On the electrodynamics of moving bodies', published in the year 1905, answers some pretty important questions on space and time. Until then, the only person who had ever touched on such notions was Newton,

who merely proclaimed that space is space and time is time. 'Tis what it is' he said, before going down to the 17th century tavern at the end of his road and drowning questions of existentialism in considerable quantities of ale. Eventually, Einstein revisited these fundamental questions and answered them with two simple ideas. Time and space are equal, and not distinct, they form the fabric of spacetime, the arena where physics occurs and each observer has his own set of coordinates (a set of rulers and a clock) and therefore he/she measures everything independently. Except for the speed of light, which he theorised is a constant for all observers. Using this as basis, he launched into the general theory of relativity. This is what provided a basis to understanding gravitational waves, so let's consider this in some detail.

Let's take Einstein's theory of special relativity to be true (as opposed to sitting and reading through a decade's worth of